



Project #62: Time-series machine learning model selection framework

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Description:

In engineering, a significant source of information stems from data collected over time. These time-series data might be event sequences, aggregated event sequences (e.g., sales data), or measurements of physical quantities (sensor data).

Machine learning workflows for time-series have several distinct phases like: windowing, feature extraction, feature selection, model fitting, model evaluation, which all need to be integrated into a cross-validated machine learning pipeline.

In order to identify generic improvements to these machine learning pipelines, an integrated high-performance computing framework needs to be implemented, which allows to test problem specific pipelines and their variations across different datasets.

Type:

Undergraduate

Outcome:

The project will extend existing frameworks for massiv parallel machine learning for the requirements of time-series machine learning, which will allow to generate statistically significant statements about potential machine learning pipeline improvements.

Prerequisites

Good Python coding and machine learning skills.

Specialisations

- Engineering Science

Categories

- Machine Learning
- Signal Processing

Supervisor

- Andreas Kempa-Liehr

Co-supervisor

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Team

- Scott Mouat
- Trisan Ethan Q Huynh

Lab

No lab has been assigned to this project